RO Membrane Fouling Early-Warning Model

Technical report — prototype v0.1.0 | 27 October 2026 | revised after API integration test

1. Objective and scope

This project packages a runnable Python command-line tool and REST API that accepts time-stamped reverse-osmosis (RO) operating data, computes a simplified normalized permeate-flow indicator, provides a 7-day model prediction and a separately labelled 14-day linear trend projection, and estimates time to a configurable cleaning threshold. It is an engineering prototype, not an operational maintenance recommendation system.

2. Data assessment and decision

Public candidate sources reviewed were (i) the U.S. Department of Energy/NREL Experimental Wave Desalination dataset and (ii) Queen's University Belfast SWRO performance workbooks. They are relevant to RO performance and operating-condition analysis, but the landing pages do not establish a long-term, labelled fouling/cleaning time series. An industrial fouling study reports SCADA analysis but data are available on request. A 422-day ultrafiltration pretreatment dataset was rejected for direct RO model training because UF and RO are different processes. Downloadable public files have not yet been fully inspected, so no public dataset is claimed as the training source.

A synthetic daily dataset is used only to make the software pipeline runnable. It varies temperature, pressure and feed TDS, applies gradual performance degradation, sensor noise and two partial cleaning resets. These assumptions are artificial and do not reproduce all physical fouling mechanisms.

3. Normalization and prediction method

The prototype estimates net driving pressure (NDP) as average feed/concentrate pressure minus permeate pressure and an approximate feed osmotic-pressure term proportional to TDS. A temperature correction factor is TCF = exp[β(T − Tref)]. The normalized-flow estimate is Qnorm = Qmeasured / TCF × NDPref / NDP. Reference temperature is 25 °C and reference NDP is 20 bar.

A post-integration review corrected a factor-of-1000 configuration error: the first-order TDS coefficient is now 0.75 bar per g/L, so 35 g/L estimates approximately 26.25 bar rather than 0.026 bar. This is only a rough seawater approximation. It ignores ionic composition, recovery-dependent concentrate salinity and detailed membrane behavior, and is not a verified ASTM D4516/vendor implementation. Plant-specific validation is required.

A Random Forest candidate predicts the normalized-flow ratio seven days ahead using lagged flow, rolling mean/slope, temperature, TDS and NDP. A persistence baseline predicts the 7-day value equals the latest value. The candidate is selected only if it improves validation MAE. Time-to-threshold is estimated separately from a recent linear trend; the model prediction and trend projection are now explicitly reported as different quantities.

4. Validation and current results

Validation is time-aware: 286 supervised rows are split chronologically into 200 training, 43 validation and 43 test rows (70%/15%/15%). Model selection uses validation data only. With the corrected normalization configuration, Random Forest validation MAE was 0.02077 normalized-flow ratio versus 0.00794 for persistence, so persistence was selected. On the held-out synthetic test segment, selected-model MAE was 0.00822 and RMSE was 0.00975.

These metrics describe this synthetic generator only and are not evidence of real-plant performance. On the sample CSV, the corrected first-order osmotic-pressure estimate is approximately 26.37 bar and calculated NDP approximately 26.24 bar. Example trend-based cleaning estimates are highly sensitive to the sample trend and should not be treated as operational advice.

RO Fouling Model — Limitations and Packaging

5. Failure modes and limitations

The candidate machine-learning model performed worse than persistence on validation, showing that a more complex model is not automatically better. Synthetic cleaning resets are difficult to infer from lagged measurements alone, and no explicit cleaning-event input is used. Trend extrapolation can fail after cleaning, feedwater changes, sensor drift, hydraulic changes, pump faults or other abrupt interventions. Public dynamic-performance experiments may not represent commercial long-term fouling. The system must not be used for plant decisions until validated on real plant data and reviewed by a qualified process engineer.

6. Software packaging and interface

The repository includes a Python package, CLI (fouling train and fouling predict), FastAPI endpoints (GET /health, POST /predict), Dockerfile, YAML configuration, versioned joblib artefact, sample CSV, data-source log and unit tests. Input validation checks required columns, timestamp parsing and duplicates, plausible ranges, positive calculated NDP, minimum row count and excessive timestamp gaps. Missing required values and suspicious units are rejected rather than silently imputed. The API integration test returned HTTP 200 for /docs and successfully processed the sample CSV through POST /predict.

7. Reproduction commands

From a Mac Terminal in the project directory: python3 -m venv .venv; source .venv/bin/activate; python -m pip install --upgrade pip; pip install -e "[dev]"; fouling train; pytest -q; fouling predict data/sample_input.csv. To run the API: uvicorn fouling.api:app --host 127.0.0.1 --port 8000, then open http://127.0.0.1:8000/docs. Docker commands are documented in README.md.

8. Next steps before operational use

Inspect the downloadable NREL and Queen’s University files at row/sheet level and determine whether any run contains meaningful chronological fouling and cleaning labels. Replace the simplified normalization with an appropriate membrane/vendor method; evaluate multiple time-aware test windows; report uncertainty for threshold crossing; test on an unseen plant file; and obtain process-engineering review. Repeat installation and testing on a clean environment.

Sources reviewed: NREL/Data.gov experimental wave desalination dataset; Queen’s University Belfast SWRO performance dataset; 2026 industrial RO fouling study; Dryad UF pretreatment dataset. See data/data_source_log.md for URLs and qualification notes.